8 Puzzle Game

```python
import heapq

goal = "123456780"

def heuristic(state):
    distance = 0

    for i in range(9):
        if state[i] != '0':
            current_row = i // 3
            current_col = i % 3

            goal_pos = int(state[i]) - 1
            goal_row = goal_pos // 3
            goal_col = goal_pos % 3

            distance += abs(current_row - goal_row)
            distance += abs(current_col - goal_col)

    return distance

def astar(start):
    pq = []
    heapq.heappush(pq, (heuristic(start), 0, start, []))

    visited = set()

    while pq:
        f, g, state, path = heapq.heappop(pq)
```

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    if state == goal:
        return path + [state]

    if state in visited:
        continue

    visited.add(state)

    blank = state.index('0')
    row = blank // 3
    col = blank % 3

    moves = [
        (-1, 0), (1, 0), (0, -1), (0, 1)
    ]

    for dr, dc in moves:
        new_row = row + dr
        new_col = col + dc

        if 0 <= new_row < 3 and 0 <= new_col < 3:
            new_pos = new_row * 3 + new_col

            temp = list(state)
            temp[blank], temp[new_pos] = temp[new_pos], temp[blank]

            new_state = ''.join(temp)

            if new_state not in visited:
```

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                    new_g = g + 1
                    new_f = new_g + heuristic(new_state)

k to add a breakpoint
                    heapq.heappush(
                        pq, (new_f, new_g, new_state, path + [state])
                    )

    return None

# Input
print("Enter the puzzle (use 0 for blank):")
start = ''.join(input().split())

# A* Search
result = astar(start)

# Output
if result:
    for i, state in enumerate(result):
        print("\nMove", i)
        for j in range(0, 9, 3):
            print(' '.join('_' if x == '0' else x
                           for x in state[j:j+3]))

    print("\nGoal State Reached")
    print("Solution Cost =", len(result) - 1)
else:
    print("No solution exists")
```

Output:

```
Enter the puzzle (use 0 for blank):
123456708

Move 0
1 2 3
4 5 6
7 _ 8

Move 1
1 2 3
4 5 6
7 8 _

Goal State Reached
Solution Cost = 1
PS C:\Users\n6787\OneDrive\Desktop\c> python -u "c:\Users\n6787

Enter the puzzle (use 0 for blank):
123406758

Move 0
1 2 3
4 _ 6
7 5 8

Move 1
1 2 3
4 5 6
7 _ 8

Move 2
1 2 3
4 5 6
7 8 _

Goal State Reached
Solution Cost = 2
```